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Inventor(s)	Honda; Naoto

Ophthalmology information processing method, control method, and ophthalmology information processing system

Abstract

A control method includes: an instruction step of transmitting an instruction to an ophthalmological imaging device to execute an imaging-related operation; an acquisition step of acquiring, from the ophthalmology imaging device, a subject's eye image taken in accordance with the instruction; and a diagnosis support step of outputting diagnosis support information for the subject's eye image. The instruction step further includes generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface.

Inventors:	Honda; Naoto (Gamagori, JP)
Applicant:	NIDEK CO., LTD. (Gamagori, JP)
Family ID:	85412353
Assignee:	NIDEK CO., LTD. (N/A, N/A)
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Primary Examiner: Huang; Frank F

Attorney, Agent or Firm: Best Mode IP Law, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of International Patent Application No. PCT/JP2021/032474 filed on Sep. 3, 2021, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to an ophthalmology information processing method, a control method, and an ophthalmology information processing system

BACKGROUND

(2) In recent years, systemization in the ophthalmology field has progressed rapidly. In such a field, a terminal device that operates a plurality of ophthalmology devices. Further, there has been known a technique for obtaining an automatic diagnosis result for each of a plurality of diseases in the subject's eye by inputting ophthalmology images into a mathematical model trained by a machine learning algorithm.

SUMMARY

(3) In one aspect of the present disclosure, an ophthalmology information processing method includes: generating, with an ophthalmology information processing system, an instruction for causing an ophthalmology imaging device configured to take a subject's eye image to execute an imaging-related operation including at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of an imaging unit of the ophthalmology imaging device, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state; transmitting, with the ophthalmology information processing system, the generated instruction to the ophthalmology imaging device; taking, with the ophthalmology imaging device, the subject's eye image in accordance with the transmitted instruction; receiving, with the ophthalmology information processing system, the subject's eye image that was taken by the ophthalmology imaging device; and outputting, with the ophthalmology information processing system, diagnosis support information for the subject's eye image. Generating the instruction further comprises generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram showing a configuration example of an ophthalmology information processing system and peripheral devices thereof.

(2) FIG. 2 is a schematic diagram showing an example of the arrangement of the ophthalmology information processing system and the peripheral devices.

(3) FIG. 3 is a schematic diagram showing an example of the arrangement of the ophthalmology information processing system and its peripheral devices.

(4) FIG. 4 is a block diagram showing the configuration of main parts of the ophthalmology information processing system and an ophthalmology imaging device.

(5) FIG. 5 is a diagram showing an example of a data structure of a patient information database.

(6) FIG. 6 is a diagram showing an example of a data structure of diagnosis support information with identification information.

(7) FIG. 7 is a diagram showing an example of a report.

(8) FIG. 8 is a diagram showing an example of a data structure of a subject's eye image with identification information.

(9) FIG. 9 is a sequence diagram showing an example of a flow of processing until a report is obtained based on a subject's eye image taken by the ophthalmology imaging device operating in a remote mode.

(10) FIG. 10 is a sequence diagram showing an example of a flow of processing until a report is obtained based on a subject's eye image taken by the ophthalmology imaging device operating in a local mode.

DESCRIPTION OF EMBODIMENTS

(11) Next, a relevant technology will be described first only for understanding the following embodiment. Medical personnel in ophthalmology and the like felt troublesome when using the ophthalmology imaging device and the image diagnostic device. For example, dedicated software customized for each of these devices had to be used. Furthermore, data registration work had to be done separately for each of these devices. Furthermore, when introducing a new image diagnostic device, a new ophthalmology imaging device adjusted to the specifications of the newly introduced image diagnostic device had to be introduced. This led to increases in human, time, and financial costs at ophthalmology related facilities.

(12) One aspect of the present disclosure aims to provide an ophthalmology information processing method, an ophthalmology information processing system, and a control method that are able to seamlessly perform operations from capturing an image of a subject's eye to supporting diagnosis based on the image of the subject's eye.

(13) According to a first aspect of the present disclosure, an ophthalmology information processing method includes: generating, with an ophthalmology information processing system, an instruction for causing an ophthalmology imaging device configured to take a subject's eye image to execute an imaging-related operation including at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of an imaging unit of the ophthalmology imaging device, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state; transmitting, with the ophthalmology information processing system, the generated instruction to the ophthalmology imaging device; taking, with the ophthalmology imaging device, the subject's eye image in accordance with the transmitted instruction; receiving, with the ophthalmology information processing system, the subject's eye image that was taken by the ophthalmology imaging device; and outputting, with the ophthalmology information processing system, diagnosis support information for the subject's eye image. Generating the instruction further comprises generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface.

(14) According to a second aspect of the present disclosure, a control method executed by an information processing device communicably connected to an ophthalmology imaging device including (i) an imaging unit that is configured to take a subject's eye image and (ii) an operation control unit that is configured to execute an imaging-related operation related to taking of the subject's eye image in accordance with an input operation directly input into the ophthalmology imaging device and is configured to execute the imaging-related operation in accordance with an instruction remotely transmitted. The method includes: an instruction step of transmitting the instruction to the ophthalmological imaging device as an operation target to execute the imaging-related operation; an acquisition step of acquiring, from the ophthalmology imaging device, the subject's eye image that was taken in accordance with the instruction; and a diagnosis support step of outputting diagnosis support information for the subject's eye image. The imaging-related operation includes at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of the imaging unit, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state. The instruction step further comprises generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface.

(15) According to a third aspect of the present disclosure, an ophthalmology information processing system communicably connected to an ophthalmology imaging device including (i) an imaging unit that is configured to take a subject's eye image and (ii) an operation control unit that is configured to execute an imaging-related operation related to taking of the subject's eye image in accordance with an input operation directly input into the ophthalmology imaging device and is

configured to execute the imaging-related operation in accordance with an instruction remotely transmitted. The system includes: at least one of (i) a circuit and (ii) a processor having a memory storing computer program code. The at least one of the circuit and the processor is configured to cause the ophthalmology information processing system to: transmit the instruction to the ophthalmological imaging device as an operation target to execute the imaging-related operation; acquire, from the ophthalmology imaging device, the subject's eye image that was taken in accordance with the instruction; and output diagnosis support information for the subject's eye image. The imaging-related operation includes at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of the imaging unit, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state. The at least one of the circuit and the processor is further configured to cause the ophthalmology information processing system to generate the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface.

(16) According to the present disclosure, it is possible to provide an ophthalmology information processing system, an ophthalmology imaging device, and a control method that are able to seamlessly perform operations from capturing an image of a subject's eye to supporting diagnosis based on the image of the subject's eye.

Embodiment

(17) Hereinafter, one embodiment of the present disclosure will be described in detail. FIG. 1 is a schematic diagram showing an example of the configuration of an ophthalmology information processing system 1 and its peripheral devices according to the present embodiment.

(18) <Overview of the Ophthalmology Information Processing System 1>

(19) The ophthalmology information processing system 1 is a system for obtaining diagnosis support information 31 and 33 based on a subject's eye image 21 captured by an ophthalmology imaging device 20. It is assumed that a business provider that designs, develops, or provides the ophthalmology imaging device 20 (hereinafter, referred to as an “ophthalmology imaging device business provider” or a “first business provider”), and a business provider that designs, develops, or provides the ophthalmology information processing system 1 (hereinafter, referred to as an “ophthalmology information processing system business provider” or a “second business provider”) are different from each other.

(20) The ophthalmology imaging device 20 includes an imaging unit (i.e., a camera) that captures an eye image 21 of a subject. The ophthalmology imaging device 20 has a function of executing an operation related to capturing the subject's eye image 21 (hereinafter, referred to as an “imaging-related operation”). A typical example of the imaging-related operation includes at least one of starting an imaging sequence, stopping the imaging sequence, switching between the left eye and the right eye, optimizing the imaging unit, alignment (adjusting the position of the device with respect to the subject's eye), capture, transfer of imaging data, initial operation associated with switching of patients, and transition to an imaging standby state. However, the imaging-related operation is not necessarily limited to the above-described examples. Optimization of the imaging unit may be, for example, focus adjustment, sensitivity adjustment, and the like. By adjusting the sensitivity, any one of light amount, gain, and exposure time can be adjusted. Further, the transition to the imaging standby state may be, for example, an operation such as switching an operation mode of the ophthalmology imaging device 20 to a power saving mode when there is no next patient after the imaging was completed.

(21) The ophthalmology imaging device 20 may have a so-called automatic alignment function. In this case, the ophthalmology imaging device 20 may further include an alignment detection unit (not shown) and a drive unit (not shown). As an example, the alignment detection unit may detect the positional relationship between the subject's eye and the imaging unit as an alignment state. Further, the drive unit may drive the imaging unit in the XYZ directions with respect to the

subject's eye. The drive section is controlled based on the detection result of the alignment state. As a result, the subject's eye and the imaging unit are adjusted to have a predetermined positional relationship suitable for imaging. Furthermore, the ophthalmology imaging device **20** may automatically capture images of the subject's eye after the automatic alignment was completed. An imaging sequence including such as an automatic alignment and an automatic imaging may be executed in one or both of a local mode and a remote mode, which will be described later.

(22) The type of the subject's eye image **21** is not necessarily restricted as long as the image **21** is an image showing a part or all of the subject's eye. The subject's eye image **21** may be, for example, a fundus image, an anterior segment image, a corneal endothelial cell image, an angle image, or the like. That is, the ophthalmology imaging device **20** may be, for example, a fundus camera, an optical coherence tomography (OCT), and a scanning fundus imaging device (SLO: Scanning Light Ophthalmoscope), an anterior segment image analysis device, a corneal endothelial cell imaging device, a gonioscopic imaging device, an automatic slit lamp imaging device, or the like. Furthermore, the type of the subject's eye image based on the imaging method is not restricted. The subject's eye image may be, for example, an SLO image, a two-dimensional OCT image, a three-dimensional OCT image, a Topo image, a slit lamp image, or the like.

(23) The ophthalmology imaging device **20** has at least an operation mode (hereinafter, referred to as a "remote mode") in which the imaging-related operation is performed in accordance with instructions from an external device (typically, the ophthalmology information processing system **1**). When the ophthalmology imaging device **20** is in the remote mode, there is no need for an operator to be present near the ophthalmology imaging device **20**. In the remote mode, the ophthalmology imaging device **20** does not need to request input of subject information including subject's identification information **24**. In the remote mode, the ophthalmology imaging device **20** transmits the captured eye image **21** to an external device (typically, the ophthalmology information processing system **1**).

(24) In addition to the remote mode, the ophthalmology imaging device **20** has an operation mode (hereinafter, referred to as a "local mode") in which the imaging-related operation is performed according to an operation input into the device itself without being remotely controlled (that is, standalone). The ophthalmology imaging device **20** requests an input of subject information including the identification information **24** during the local mode. During the local mode, the ophthalmology imaging device **20** may or may not transmit the subject's eye image **22** to an external device (typically, the ophthalmology information processing system **1**). The subject's eye image **22** is an image in which the subject's eye image **21** and the subject's identification information **24** are associated with each other.

(25) In the following, it is assumed that the ophthalmology imaging device **20** has the remote mode and the local mode. The ophthalmology imaging device **20** will be described as being capable of switching between the remote mode and the local mode. In other words, the ophthalmology imaging device **20** can perform the imaging-related operation in accordance with operations input into the device. Furthermore, it is assumed that the ophthalmology imaging device **20** is capable of performing the imaging-related operation in accordance with instructions from an external device. Note that if the ophthalmology imaging device **20** having the both modes is manufactured, there is no need to provide separated product lineups for the ophthalmology imaging device **20**.

(26) The ophthalmology information processing system **1** includes at least the ophthalmology information processing device **10** and a diagnosis support system **30**, as shown in FIG. **1**. The business provider that designs, develops, or provides the ophthalmology information processing device **10** and the business provider that designs, develops, or provides the diagnosis support system **30** may be the same or different. In other words, the ophthalmology information processing system provider may be a plurality of providers. Moreover, each function of the ophthalmology information processing device **10** and the diagnosis support system **30** may be integrated and implemented in a single information processing device. Alternatively, each function of the

ophthalmology information processing device **10** and the diagnosis support system **30** may be distributed and implemented in two or more information processing devices. For example, the ophthalmology information processing device **10** may also have the functions of the diagnosis support system **30**. Further, for example, the functions of the diagnosis support system **30** may be distributed to a plurality of medical facilities.

(27) The ophthalmology information processing device **10** is communicatively connected to one or more ophthalmology imaging devices **20**. The ophthalmology information processing device **10** gives instructions to each of the ophthalmology imaging devices **20** that is operating in the remote mode. Specifically, the ophthalmology information processing device **10** instructs each of the ophthalmology imaging devices **20** that is operating in the remote mode to perform the imaging-related operation. Then, the ophthalmology information processing device **10** acquires the subject's eye image **21** captured by the instructed ophthalmology imaging device **20** therefrom. At this time, one of the characteristics of the ophthalmology information processing system **1** is that a dedicated software interface provided by an ophthalmology imaging device provider is used for communication with the ophthalmology imaging device **20**. The form of the software interface is not restricted. The software interface may typically be an API (Application Programming Interface) or equivalent commands. Further, the software interface may be provided as an SDK (Software Development Kit) including an API or equivalent commands. By using the dedicated software interface, the design and development of software required for interacting with the ophthalmology imaging device **20** can be easily realized at low cost. Furthermore, by using the dedicated software interface, it is useful in that the design and development of software can be realized with a high degree of freedom without subject to restrictions such as a specification of the ophthalmology imaging device **20**. For example, the ophthalmology information processing system provider can directly control the ophthalmology imaging device **20**. By doing so, the degree of freedom in realizing the optimization design of the entire operation including the ophthalmology information processing system **1** and in the design and development of the operation interface, etc. can be increased. It is even more useful when working with a plurality of ophthalmology imaging devices **20** having different specifications. It should be noted that conventional control software used to control an ophthalmology imaging device was usually only installed and used, and was not able to be flexibly improved. In other words, it was difficult to perform design and development with a high degree of freedom as described above.

(28) The ophthalmology information processing device **10** transmits the subject's eye images **21** and **22** acquired from the ophthalmology imaging device **20** to the diagnosis support system **30** that is communicably connected to the processing device **10**. The diagnosis support system **30** is a system for providing diagnosis support. The diagnosis support system **30** receives the image of the subject's eye **21**, as input, acquired from the ophthalmology information processing device **10**. The diagnosis support system **30** then outputs diagnosis support information **31** for the subject's eye image **21**. Furthermore, the diagnosis support system **30** receives the subject's eye image **22** acquired from the ophthalmology information processing device **10** as input. The diagnosis support system **30** then outputs diagnosis support information **33** for the subject's eye image **22**. Typically, one or more computers in the diagnosis support system **30** performs the diagnosis support. However, one or more doctors may perform the diagnosis support. The process on the subject's eye images **21** and **22** is not restricted. For example, such a process may include analysis processing, processing using artificial intelligence, or a combination thereof.

(29) The diagnosis support system **30** transmits the output diagnosis support information **31** and **33** to the ophthalmology information processing device **10**.

(30) The ophthalmology information processing device **10** may generate diagnosis support information **32**. The diagnosis support information **32** is associated with (i) the diagnosis support information **31** acquired from the diagnosis support system **30** and (ii) the identification information **54** of a subject whose subject's eye image, which is the source of the diagnosis support

information **31**, was taken. Here, the source of the identification information **54** is not necessarily the ophthalmology imaging device **20**. That is, one of the features of the ophthalmology information processing system **1** is that the diagnosis support information **32** associated with the identification information **54** can be generated without depending on the ophthalmology imaging device **20**. Accordingly, it is not necessary to register and manage identification information of the subject at the ophthalmology imaging device **20**. Furthermore, the identification information does not need to be linked between the ophthalmology imaging device **20** and the ophthalmology information processing system **1**. Therefore, it is possible to reduce the complexity associated with registering and managing the identification information of the subject. Note that conventionally the identification information of a subject had to be registered at the ophthalmology imaging device. Furthermore, even in a system that handled images of the subject's eye taken by an ophthalmology imaging device, the identification information of the subject was usually registered for the system, which made data registration and data management complicated.

(31) The source of the identification information **54** is, for example, but not limited to, an electronic medical record system **50** communicatively connected to the ophthalmology information processing device **10**. The electronic medical record system **50** is a system having a database that manages electronic medical records including patient information. Note that the managing operation manner in the electronic medical record system **50** may be different from that of the ophthalmology information processing system **1**.

(32) The ophthalmology information processing device **10** may generate a report **41** including the diagnosis support information **32** or the diagnosis support information **33**. The report **41** may include the images **21** and **22** of the subject's eye. The ophthalmology information processing device **10** may transmit the generated report **41** to the electronic medical record system **50** via a communication network N. The report **41** may be stored in the electronic medical record system **50**. The report **41** may be transmitted to an output device **40** directly from the electronic medical record system **50** or via the ophthalmology information processing system **1**. Typical examples of the output device **40** are a PC (Personal Computer) used by a medical professional or a patient, an information processing terminal device such as a tablet terminal device, a printer, etc. However, other devices may serve as the output device **40** as long as the devices are capable of displaying, printing, or the like, the report **41**.

(33) The ophthalmology information processing system **1** includes the functions as described above. Therefore, the ophthalmology information processing system **1** is a system that is functionally different from a conventional information processing terminal device equipped with control software for controlling an ophthalmology imaging device and an image viewer for viewing images of the subject's eye taken by the ophthalmology imaging device. According to the ophthalmology information processing system **1**, it is possible to seamlessly perform operations from taking the subject's eye image to supporting diagnosis based on the subject's eye image. In other words, it is possible to reduce the workload of ophthalmology medical personnel.

(34) Furthermore, as described above, the ophthalmology information processing system **1** provided by the ophthalmology information processing system provider uses the dedicated software provided by the ophthalmological imaging device provider when communicating with the ophthalmology imaging device **20** provided by the ophthalmological imaging device provider. For example, the ophthalmology information processing system **1** generates instructions in line with specifications of the software interface. Then, the ophthalmology information processing system **1** gives the instructions to the ophthalmology imaging device **20**. Further, the ophthalmology imaging device **20** executes the imaging-related operations based on the instructions. Therefore, the ophthalmology information processing system **1** provided by the ophthalmology information processing system provider can perform a series of processes from taking the subject's eye image **21** by the ophthalmology imaging device **20** to obtaining the diagnosis support information **31**, **33** based on the subject's eye image **21** without subject to constraints by other specifications,

environment, etc. of the ophthalmology imaging device **20** provided by the ophthalmology imaging device provider. Further, for the ophthalmology imaging device provider, there is no need to customize the ophthalmology imaging device **20** to meet the specifications of the ophthalmology information processing system **1** provided by the ophthalmology information processing system provider. Therefore, the ophthalmology imaging device **20** can be operated based on instructions from the ophthalmology information processing system **1**.

Example of Arrangement of Devices

(35) One example of the arrangement of devices will be described with reference to FIGS. **2** and **3**. A typical example of the location where the ophthalmology information processing device **10** is located in a medical related facility M, as shown in FIG. **2**. A typical example of the medical related facility M is a hospital. However, the medical related facility M may be a clinic, a testing facility, a pharmacy, a drug store, or an ophthalmology related store. The ophthalmology information processing device **10** is connected to the communication network N including a LAN (Local Area Network) installed in, e.g., the medical related facility M.

(36) A typical example of the location of the diagnosis support system **30** is, as shown in FIG. **2**, a facility different from the medical related facility M where the ophthalmology information processing device **10** is located. In this case, the ophthalmology information processing device **10** is connected to the diagnosis support system **30** via, for example, a VPN (Virtual Private Network) on the Internet. The diagnosis support system **30** may be a system configured by a server or the like built on a cloud. Note that the diagnosis support system **30** may be placed in the medical related facility M where the ophthalmology information processing device **10** is located. In this case, the ophthalmology information processing device **10** and the diagnosis support system **30** are connected to each other via the LAN installed in the medical related facility M.

(37) A typical example of the location for the output device **40** is the medical related facility M where the ophthalmology information processing device **10** is placed, as shown in FIG. **2**. In this case, the output device **40** is connected to the communication network N including the LAN installed in the medical related facility M, for example. Alternatively, the output device **40** may be placed in a facility different from the medical related facility M where the ophthalmology information processing device **10** is placed. In this case, the communication network N to which the output device **40** is connected may include a VPN on the Internet or the like. Furthermore, the output device **40** may be a portable device and may not be fixedly placed.

(38) A typical example of the location for the electronic medical record system **50** is the medical related facility M where the ophthalmology information processing device **10** is located. In this case, the electronic medical record system **50** is connected to the communication network N including the LAN installed in the medical related facility M, for example.

(39) The ophthalmology imaging device **20** may be located in a variety of locations. A first typical example of the location for the ophthalmology imaging device **20** is the medical related facility M, as shown in FIG. **2**. The ophthalmology imaging device **20** located in the medical related facility M is connected to the ophthalmology information processing device **10** via the LAN installed in the medical related facility M. A plurality of ophthalmology information processing devices **10** may be placed in the medical related facility M. In the case of the example shown in FIG. **2**, the subject's eye images **21** and **22** are taken, the diagnosis support information **31** and **33** are acquired, and the report **41** is output in the medical related facility M. The operators of the ophthalmology imaging device **20**, the ophthalmology information processing device **10**, and the output device **40** may be the same person.

(40) A second typical example of the location for the ophthalmology imaging device **20** is a medical related facility L that is different from the medical related facility M, as shown in FIG. **3**. The medical related facility L may be a clinic, a testing facility, a pharmacy, a drug store, or an ophthalmology-related store. The ophthalmology imaging device **20** located in the medical related facility L is connected to the ophthalmology information processing device **10** via, for example, a

VPN on the Internet. As shown in FIG. 3, each of the plurality of ophthalmology imaging devices **20** may be placed in a mutually different one of the medical related facilities **L**. In the example shown in FIG. 3, each of the subject's eye images **21** and **22** taken at one or more medical related facilities **L** is collected at one medical related facility **M**. Then, the medical related facility **M** acquires the diagnosis support information **31** and **33** and outputs the report **41**. In the case of the example shown in FIG. 3, the operator of the ophthalmology imaging device **20** and the operators of the ophthalmology information processing device **10** and the output device **40** are usually different from each other. It is assumed that the ophthalmology imaging devices **20** placed in the respective medical related facilities **L** may have different specifications, environments, and the like. Therefore, in the example shown in FIG. 3, it is preferable that the ophthalmology information processing system **1** utilizes a dedicated software interface when communicating with the ophthalmology imaging device **20**.

(41) <Main Configuration of the Ophthalmology Information Processing System **1**>

(42) FIG. 4 is a block diagram showing the main configuration of the ophthalmology information processing system **1** and the ophthalmology imaging device **20**. As described above, the ophthalmology information processing system **1** typically includes the ophthalmology information processing device **10** and the diagnosis support system **30**.

(43) (Ophthalmology Information Processing Device **10**)

(44) The ophthalmology information processing device **10** includes an input unit **107**, a control unit **100**, and a storage unit **110**.

(45) The input unit **107** is a device that accepts input operations input into the ophthalmology information processing device **10**. For example, the input unit **107** may be a keyboard, a mouse, a touch panel, and a microphone. When the input unit **107** is a touch panel, the input unit **107** may also have an information display function. The input unit **107** outputs a signal corresponding to the received input operation to the control unit **100**.

(46) The storage unit **110** is a memory that stores various data and various software used by the ophthalmology information processing device **10**. Alternatively, the storage unit **110** may be a storage device connected to the ophthalmology information processing device **10**.

(47) The storage unit **110** may include an image storage unit **1101**, a diagnosis support information storage unit **1102**, and a report storage unit **1103**. The image storage unit **1101** stores the subject's eye images **21** and **22**. The diagnosis support information storage unit **1102** stores the diagnosis support information **31** and **33**. The report storage unit **1103** stores the report **41**.

(48) The control unit **100** comprehensively controls each function of the ophthalmology information processing device **10**. The control unit **100** is, for example, a processor such as a CPU (Central Processing Unit). For example, the control unit **100** reads out a program for performing each function from the storage unit **110** and sends the program in a RAM (Random Access Memory). That is, the control unit **100** includes at least a communication control unit **101**. The communication control unit **101** includes an operation instruction unit **1011**, an image acquisition unit **1012**, and a software interface (hereinafter, referred to as "software IF") **1013**. Further, the control unit **100** may include an operation target selection unit **103**, a subject information acquisition unit **104**, an association unit **105**, and a report generation unit **106**.

(49) Programs for performing functions of the operation instruction unit **1011**, the image acquisition unit **1012**, the subject information acquisition unit **104**, the association unit **105**, and the report generation unit **106** are provided by the ophthalmology information processing system provider. On the other hand, the software IF **1013** is provided from the ophthalmology imaging device provider directly or via an intermediary provider to the ophthalmology information processing system provider.

(50) The operation target selection unit **103** selects an ophthalmology imaging device **20**, as an operation target, from among the plurality of ophthalmology imaging devices **20**. The ophthalmology imaging device **20**, as an operation target, is an ophthalmology imaging device that

is configured to operate in a remote mode. Note that if only one predetermined ophthalmology imaging device **20** exists as an operation target, there is no need for the operation target selection unit **103** to select the ophthalmology imaging device **20**. Alternatively, the operation target selection unit **103** may select the ophthalmology imaging device **20** as an operation target, in accordance with the input operation received by the input unit **107**. Furthermore, the operation target selection unit **103** may automatically select the ophthalmology imaging device **20**, as an operation target, based on a predetermined rule.

(51) The operation instruction unit **1011** instructs the ophthalmology imaging device **20** to perform the imaging-related operation. It is preferable that the instructions meet the specifications of the software IF **1013** (that is, the instructions are generated via the software IF **1013**). Specifically, it is preferable that the operation instruction unit **1011** transmits a command **11** that meets the specifications of the software IF **1013** to cause the ophthalmology imaging device **20** to perform the imaging-related operation.

(52) The software IF **1013** instructs the ophthalmology imaging device **20** to perform the imaging-related operation. The software IF **1013** is an aggregation of dedicated software interfaces for acquiring the subject's eye images **21** and **22**. The software IF **1013** is typically an aggregation of APIs. For example, the software IF **1013** is distributed or downloaded to the ophthalmology information processing device **10** from a server (not shown) that provides the software IF **1013**. Furthermore, the software IF **1013** is installed in the ophthalmology information processing device **10**. Further, as an example, the software IF **1013** stored on a storage medium such as a USB (Universal Serial Bus) memory may be read and installed in the ophthalmology information processing device **10**. In this way, the method of providing the software IF **1013** is not restricted.

(53) The timing at which the operation instruction unit **1011** instructs the ophthalmology imaging device **20** to perform the imaging-related operations is not restricted. The timing may be, for example, a timing immediately after the ophthalmology imaging device **20** was selected by the operation target selection unit **103**. Alternatively, the timing may be a timing upon receiving an input operation by the input unit **107**. Furthermore, the timing may be determined based on a predetermined rule.

(54) The image acquisition unit **1012** acquires the subject's eye image **21** that was taken by the ophthalmology imaging device **20** which was operating in the remote mode according to instructions from the operation instruction unit **1011**. It is preferable that the image acquisition unit **1012** acquires the subject's eye image **21** via the software IF **1013**.

(55) The image acquisition unit **1012** transmits the subject's eye image **21** acquired from the ophthalmology imaging device **20** to the diagnosis support system **30**. The transmission timing of the subject's eye image **21** is not restricted. The transmission timing of the subject's eye image **21** may be, for example, a timing immediately after the image acquisition unit **1012** acquired the subject's eye image **21**. Further, the transmission timing may be a timing upon receiving an input operation by the input unit **107**. Furthermore, the timing may be a timing determined based on a predetermined rule.

(56) The image acquisition unit **1012** may store the subject's eye image **21** acquired from the ophthalmology imaging device **20** in the image storage unit **1101**. The storage timing of the subject's eye image **21** is not restricted. For example, the storage timing may be a timing immediately after the image acquisition unit **1012** acquired the subject's eye image **21**. Alternatively, the storage timing may be a timing before the image acquisition unit **1012** transmits the subject's eye image **21** to the diagnosis support system **30**. Furthermore, the storage timing may be a timing after the image acquisition unit **1012** transmitted the subject's eye image **21** to the diagnosis support system **30**. Yet furthermore, the storage timing may be a timing determined based on a predetermined rule.

(57) The image acquisition unit **1012** may acquire the subject's eye image **22** that was taken by the ophthalmology imaging device **20** which was operating in the local mode. Also in this case, the

image acquisition unit **1012** preferably acquires the subject's eye image **22** via the software IF **1013**. The image acquisition unit **1012** may transmit the acquired subject's eye image **22** to the diagnosis support system **30**. The transmission timing of the subject's eye image **22** is not restricted. The transmission timing may be, for example, a timing immediately after the image acquisition unit **1012** acquired the subject's eye image **22**. Further, the transmission timing may be a timing upon receiving an input operation by the input unit **107**, or a timing determined based on a predetermined rule.

(58) The image acquisition unit **1012** may store the acquired subject's eye image **22** in the image storage unit **1101**. The storage timing of the subject's eye image **22** is not restricted. The storage timing may be various timings. For example, the subject's eye image **22** may be stored immediately after the image acquisition unit **1012** acquired the subject's eye image **22**. Alternatively, the subject's eye image **22** may be stored before the image acquisition unit **1012** transmits the subject's eye image **22** to the diagnosis support system **30**. Furthermore, the subject's eye image **22** may be stored after the image acquisition unit **1012** transmitted the subject's eye image **22** to the diagnosis support system **30**, or the subject's eye image **22** may be stored at a timing determined based on a predetermined rule.

(59) The subject information acquisition unit **104** acquires subject information **53** that includes at least the identification information **54** of the subject whose eye image **21** is to be taken. The subject information acquisition unit **104** may acquire the subject information **53** in response to an input operation received by the input unit **107**. The acquisition timing of the subject information **53** is not restricted. The acquisition timing may be, for example, a timing before the image acquisition unit **1012** transmits the subject's eye image **21** to the diagnosis support system **30**. Further, the acquisition timing may be a timing after the image acquisition unit **1012** transmitted the subject's eye image **21** to the diagnosis support system **30**, or may be a timing determined according to an input operation received by the input unit **107**.

(60) The source of the subject information **53** is, for example, a patient information database **501** included in the electronic medical record system **50**. The patient information database **501** is a database that stores patient information.

(61) FIG. 5 is a diagram showing an example of a data structure of the patient information database **501**. The patient information database **501** is shown in FIG. 5. The patient information database **501** may include information such as name, age, gender, medical history, previous examination date, hospital visit history, etc. as the subject information **53**, using a patient ID, which is the identification information **54**, as a key. Furthermore, the patient information database **501** may further include information such as past subject's eye images **21** and **22**, the diagnosis support information **31** and **33**, and findings. In addition to the patient ID as the identification information **54**, the subject information acquisition unit **104** may acquire, from the patient information database **501**, respective pieces of the above-mentioned information associated with the patient ID.

(62) The association unit **105** acquires the diagnosis support information **31** for the subject's eye image **21** for the subject from the diagnosis support system **30**. The association unit **105** generates the diagnosis support information **32** attached with the identification information. The diagnosis support information **32** is generated by associating the acquired diagnosis support information **31** with the subject information **53** (including at least the identification information **54**). This subject information **53** is information about the subject whose eye image **21** was taken. The diagnosis support information **31** is originated from the subject's eye image. The information **53** is acquired by the subject information acquisition unit **104**. The generation timing of the diagnosis support information **32** is not restricted. The generation timing of the diagnosis support information **32** may be, for example, a timing immediately after acquiring both the diagnosis support information **31** and the subject information **53**. Alternatively, the diagnosis support information **32** may be generated upon receiving an input operation by the input unit **107**.

(63) As described above, the identification information **54** is acquired from the patient information

database **501** of the electronic medical record system **50**. That is, the ophthalmology information processing system **1** can generate the diagnosis support information **32** associated with the identification information **54** without depending on the ophthalmology imaging device **20**. Therefore, it is not necessary to register and manage identification information of subjects in the ophthalmology imaging device **20**. Furthermore, the identification information does not need to be linked between the ophthalmology imaging device **20** and the ophthalmology information processing system **1**. For the reasons described above, using the ophthalmology information processing system **1**, it is possible to reduce the complexity involved in registering and managing the identification information of subjects.

(64) FIG. **6** is a diagram showing an example of the data structure of the diagnosis support information **32** attached with the identification information. In the example shown in FIG. **6**, the diagnosis support information **31** is expressed in a single folder. It should be understood that one or more files (not shown) related to the diagnosis support information **31** are presented in the folder. In the example shown in FIG. **6**, only the identification information **54** is associated with the diagnosis support information **31**, but the present disclosure is not necessarily limited to this. For example, in addition to the identification information **54**, the association unit **105** may generate the diagnosis support information **32** by associating each piece of information acquired by the subject information acquisition unit **104** from the patient information database **501** with the diagnosis support information **31**. The information acquired by the subject information acquisition unit **104** includes, for example, name, age, gender, medical history, previous examination date, hospital visit history, past subject's eye images **21** and **22**, and findings.

(65) The association unit **105** may store the diagnosis support information **31** and **32** in the diagnosis support information storage unit **1102**. Alternatively, the association unit **105** may store the diagnosis support information **31** and **32** in the patient information database **501**.

(66) The report generation unit **106** generates the report **41** including the diagnosis support information **32**. The report generation unit **106** may acquire the diagnosis support information **32** from the association unit **105**. Alternatively, the report generation unit **106** may read the diagnosis support information **32** from the diagnosis support information storage unit **1102**.

(67) Further, the report generation unit **106** generates the report **41** including the diagnosis support information **33** for the subject's eye image **22**. The report generation unit **106** may acquire the diagnosis support information **33** from the diagnosis support system **30**. Alternatively, the report generation unit **106** may read the diagnosis support information **33** from the diagnosis support information storage unit **1102**.

(68) The generation timing of the report **41** is not restricted, and may be, for example, a timing immediately after the report generation unit **106** acquired the diagnosis support information **32** and **33**. Alternatively, the report **41** may be generated upon receiving an input operation by the input unit **107**. The report generation unit **106** may store the generated report **41** in the report storage unit **1103**. The report generation unit **106** may store the acquired diagnosis support information **32** and **33** in the diagnosis support information storage unit **1102**.

(69) A reader of the report **41** may be medical personnel, a subject, or both. Therefore, the report generation unit **106** may generate the report **41** in a displaying format according to the reader. For example, the report **41** for medical personnel may highly detailed or may use ophthalmology terminology. On the other hand, the report **41** for subjects may be less detailed or may not use ophthalmology terminology. Alternatively, the report generation unit **106** may determine the display format of the report **41** according to an input operation received by the input unit **107**.

(70) The report generation unit **106** transmits the generated report **41** to the electronic medical record system **50** via the communication network **N**. The transmission timing of the report **41** is not limited. For example, the report generation unit **106** may transmit the report **41** at a timing immediately after generating the report **41**. Further, the report **41** may be transmitted at a timing upon receiving an input operation by the input unit **107**. Note that the electronic medical record

system **50** may transmit the report **41** to the output device **40** directly or via the ophthalmology information processing system **1**. The output device **40** as an output destination may be determined in advance. Alternatively, the output device **40** as an output destination may be determined according to an input operation received by the electronic medical record system **50**. Furthermore, the output device **40** as an output destination may be determined according to an input operation received by the ophthalmology information processing system **1**. The method for transmitting to the output device **40** is not restricted. For example, the report **40** may be transmitted using a file transfer system. Alternatively, the report **40** may be transmitted as an attachment to a general-purpose email. The file format of the report **41** is not limited, and may be, for example, PDF (Portable Document Format).

(71) The report generation unit **106** may directly transmit the generated report **41** to the output device **40**. Further, the report generation unit **106** may store the generated report **41** in the report storage unit **1103**.

(72) FIG. 7 is a diagram showing an example of the report **41**. The report **41** shown in FIG. 7 shows one example including the diagnosis support information **32**. The report **41** includes, for example, a subject information display area **411** and a diagnosis support information display area **412**. Further, the report **41** may further include, for example, an image display area **413** and a findings display area **414**.

(73) In the subject information display area **411**, all or a part of the subject information **53** associated with the diagnosis support information **32** is displayed. The subject information display area **411** may display, for example, the identification information **54**, name, gender, age, medical history, previous examination date, hospital visit history, and the like. For example, if the subject needs to be prevented from being identified, the name may not be displayed.

(74) In the diagnosis support information display area **412**, all or a part of the text indicating the diagnosis support information included in the diagnosis support information **32** is displayed. FIG. 7 shows one example of symptoms of diabetic retinopathy and severity classification. As an example, texts such as “macular edema” and “simple retinopathy” are displayed.

(75) The image display area **413** displays, for example, the subject's eye image **21** (which may be processed by, e.g., highlighting an abnormal location) included in the diagnosis support information **32**. Further, in the image display area **413**, the image analysis result **35** of the subject's eye image **21**, etc. are displayed. The image display area **413** may display, for example, a past subject's eye image **21** of the same subject and image analysis results of the subject's eye images **21**. By displaying the past subject's eye images **21**, medical personnel and the patient can easily check the progress of symptoms.

(76) The findings display area **414** is, for example, an area for medical personnel to note their findings therein. The report **41** with the noted findings may be transmitted to an information processing terminal device used by the subject. At this time, the report **41** is transmitted from the information processing terminal device used by the doctor directly or via the ophthalmology information processing system **1** to the information processing terminal device of the subject.

(77) (Diagnosis Support System **30**)

(78) The diagnosis support system **30** includes a diagnosis support unit **301**. The diagnosis support unit **301** receives, as an input, the subject's eye image **21** acquired from the ophthalmology information processing device **10** and outputs, as an output, diagnosis support information **31** on the subject's eye image **21**. Similarly, the diagnosis support unit **301** receives as an input, the subject's eye image **22** acquired from the ophthalmology information processing device **10** and outputs, as an output, diagnosis support information **33** on the subject's eye image **22**. The processing contents by the diagnosis support unit **301** may be image analysis or processing using artificial intelligence. When using artificial intelligence, the diagnosis support unit **302** may use a diagnosis model generated by machine-learning the correlation between the subject's eye image and the diagnosis support information.

(79) A program for realizing functions of the diagnosis support unit **301** is provided by the ophthalmology information processing system provider.

(80) The diagnosis support information **31**, **33** may be formed of one or more electronic files including images, texts, numerical data, and combinations thereof. The images included in the diagnosis support information **31** and **33** include the subject's eye images **21** and **22** themselves, and images that have been processed by, e.g., highlighting abnormalities in the subject's eye images **21** and **22**. However, the images included in the diagnosis support information **31** and **33** are not necessarily limited to these. The texts included in the diagnosis support information **31** and **33** may be a text representing an image analysis result, a text representing diagnosis support information, a text representing a treatment method, and the like. However, it is not necessarily limited to these. Specific examples of the text included in the diagnosis support information **31** and **33** include disease name, symptoms, degree of progress of symptoms, and therapeutic drugs. However, it is not necessarily limited to these.

(81) As an example, it is assumed that the subject's eye images **21** and **22** are retinal images, and macular edema is recognized through diagnosis support. In this case, the diagnosis support unit **301** may output the diagnosis support information **31** and **33** including an image clearly showing the location of macular edema and a text such as “MACULAR EDEMA PRESENT” and “SIMPLE RETINOPATHY”. Accordingly, it is possible to create the report **41** as shown in FIG. 7, for example.

(82) <Main Configuration of the Ophthalmology Imaging Device **20**>

(83) Returning back to FIG. 4, the main configuration of the ophthalmology imaging device **20** will be described. The ophthalmology imaging device **20** is shown in FIG. 4. The main component of the ophthalmology imaging device **20** includes, for example, a control unit **200**, an imaging unit **206**, an input unit **207**, and a storage unit **210**.

(84) The imaging unit **206** is a camera for taking the subject's eye image **21**. The imaging unit **206** is controlled by an operation control unit **202**, which will be described later. The imaging unit **206** may add EXIF (Exchangeable Image File Format) information such as imaging date and time and position information to the subject's eye image **21**.

(85) The input unit **207** is a device that accepts input operations into the ophthalmology imaging device **20**. A typical example of the input operation is an operation for causing the ophthalmology imaging device **20** operating in the local mode to perform the imaging-related operation. Furthermore, a typical example of the input operation is an operation for inputting or selecting the subject information **23**.

(86) The input unit **207** may be, for example, a touch panel, a button, or a switch mounted in the ophthalmology imaging device **20**. Further, the input unit **207** may be a mouse and a keyboard of an information terminal device connected to the ophthalmology imaging device **20**. When the input unit **207** is a touch panel, the input unit **207** may also have an information display function. The input unit **207** outputs a signal corresponding to the received input operation to the control unit **200**.

(87) The storage unit **210** is a memory that stores various data and various software used by the ophthalmology imaging device **20**. Alternatively, the storage unit **210** may be a storage device connected to the ophthalmology imaging device **20**. Furthermore, the storage unit **210** may include an image storage unit **2101** and a subject information storage unit **2102**. The image storage unit **2101** stores the subject's eye images **21**. The subject information storage unit **2102** stores the subject information **23** including the identification information **24** of the subject. Typical examples of information included in the subject information **23** are the identification information **24**, name, date of birth, gender, and the like. The code system of the identification information **24** may be different from that of the patient ID managed by the electronic medical record system **50**.

(88) The control unit **200** centrally controls each function of the ophthalmology imaging device **20**. The control unit **200** is, for example, a processor such as a CPU. For example, the control unit **200** reads out a program for realizing each function from the storage unit **210** and expands the program

to the RAM. The control unit **200** includes a mode switching unit **201**, an operation control unit **202**, a subject information receiving unit **203**, and an association unit **205** as a result of the expansion.

(89) The mode switching unit **201** switches the operation mode of the ophthalmology imaging device **20**. Specifically, the mode switching unit **201** switches between the local mode and the remote mode. The mode switching unit **201** typically switches the mode according to an input operation received by the input unit **207**. Alternatively, the mode switching unit **201** may switch the mode according to instructions from an external device. Further, the mode switching unit **201** may switch the mode according to a predetermined rule.

(90) The operation control unit **202** controls the imaging unit **206** and executes the imaging-related operation. The operation control unit **202** switches between the remote mode and the local mode.

(91) The processing contents of the operation control unit **202** in the remote mode are as follows. The operation control unit **202** executes the imaging-related operations according to instructions from the ophthalmology information processing system **1** (specifically, the ophthalmology information processing device **10**). As such, the operation control unit **202** causes the imaging unit **206** to take the subject's eye image **21**. If a command is transmitted from the ophthalmology information processing device **10** via the software IF **1013**, the operation control unit **202** may receive the command. The operation control unit **202** may then execute the imaging-related operation according to the content of the received command. The operation control unit **202** acquires the subject's eye image **21** taken by the imaging unit **206**. The operation control unit **202** may add related information to the acquired subject's eye image **21**. The related information may be, for example, identification information of the imaging unit **206**. The operation control unit **202** transmits the acquired eye image **21** to the ophthalmology information processing system **1** (specifically, the ophthalmology information processing device **10**). Here, it should be noted that the subject's eye image **21** is not associated with the identification information of the subject. The transmitted subject's eye image **21** is acquired by the image acquisition unit **1012** via the software IF **1013** in the ophthalmology information processing device **10**. Note that the operation control unit **202** may store the subject's eye image **21** in the image storage unit **2101**. The subject's eye image **21** stored in the image storage unit **2101** may be output (displayed, printed, etc.) to an output device (not shown).

(92) The processing contents of the operation control unit **202** in the local mode are as follows. The operation control unit **202** follows input operations from the input unit **207**. Therefore, the operation control unit **202** causes the imaging unit **206** to take the subject's eye image **21** by executing the imaging-related operation. The operation control unit **202** acquires the subject's eye image **21** taken by the imaging unit **206**. The operation control unit **202** may add related information to the acquired subject's eye image **21**. The related information may be, for example, identification information of the imaging unit **206**. The operation control unit **202** transmits the subject's eye image **21** taken by the imaging unit **206** to the association unit **205**. The operation control unit **202** may acquire the subject's eye image **22** from the association unit **205**. The subject's eye image **22** is generated by the association unit **205** based on the subject's eye image **21**. Further, the subject's eye image **22** is associated with the patient information **23** including the identification information **24**. The operation control unit **202** may transmit the acquired eye image **22** to the ophthalmology information processing system **1** (specifically, the ophthalmology information processing device **10**). Note that the operation control unit **202** may store the subject's eye image **22** in the image storage unit **2101**. the subject's eye image **22** stored in the image storage unit **2101** may be output (displayed, printed, etc.) to an output device (not shown).

(93) The subject information receiving unit **203** accepts input or selection of the subject information **23** in accordance with an input operation from the input unit **207** during the local mode. The person performing the input operation may be the subject. Alternatively, the person performing the input operation may be a medical staff. The subject information **23** may be newly

input. Alternatively, the subject information **23** may be read from the subject information storage unit **2102** and selected. The subject information **23** includes at least the identification information **24** of the subject. The identification information **24** may be newly issued. Alternatively, the identification information **24** may be read from the subject information storage unit **2102** and selected.

(94) In the remote mode, the subject information receiving unit **203** may operate in the same manner as in the local mode. Alternatively, the subject information receiving unit **203** may not accept input or selection of the subject information **23**. In the latter case, the subject information receiving unit **203** may display, on the input unit **207** which also has a display function, a message that the input or selection of the subject information **23** is not accepted. Alternatively, the subject information receiving unit **203** may cause a display unit (not shown) to display such a message.

(95) In the remote mode, the operation control unit **202** may perform imaging-related operations regardless of whether the subject information **23** is input. That is, in the remote mode, even if the subject information receiving unit **203** receives the input or selection of the subject information **23**, the operation control unit **202** may execute the imaging-related operation instructed under the remote mode.

(96) The association unit **205** acquires the subject's eye image **21** taken by the imaging unit **206** from the operation control unit **202** during the local mode. The association unit **205** generates the subject's eye image **22** by associating the acquired subject's eye image **21** with the subject information **23** that includes at least the identification information **24** and is received by the subject information receiving unit **203**. FIG. **8** is a diagram showing an example of the data structure of the subject's eye image **22** attached with the identification information. In the example shown in FIG. **8**, the subject's eye image **21** is expressed as one folder. It should be understood that one or more files (not shown) related to the subject's eye image **21** are present in the folder. In the example shown in FIG. **8**, only the identification information **24** is associated with the subject's eye image **21**, which is not necessarily limited to this. For example, in addition to the identification information **24**, the association unit **205** may associate the subject information **23** (such as name) other than the identification information **24** with the subject's eye image **21**. That is, the association unit **205** may generate the subject's eye image **22**.

(97) The association unit **205** may operate in the remote mode in the same manner as in the local mode. Furthermore, the association unit **205** may stop operating.

(98) The association unit **205** may store the subject's eye image **22** in the image storage unit **2101**.

(99) <Processing Flow>

(100) FIG. **9** is a sequence diagram showing an example of a flow of processing until the report **41** is obtained based on the subject's eye image **21** taken by the ophthalmology imaging device **20** operating in the remote mode. Each process in the ophthalmology information processing system **1** is executed and controlled by one or more information processing devices. At least one information processing device may be communicably connected to the ophthalmology imaging device **20**.

(101) In the ophthalmology imaging device **20**, the mode switching unit **201** switches the mode to the remote mode in response to an input operation received by the input unit **207** (step S1). Note that step S1 may be executed at the latest before step S3 is executed.

(102) In the ophthalmology information processing device **10**, the operation target selection unit **103** selects the ophthalmology imaging device **20** that is a target for operation (step S2). The operation instruction unit **1011** instructs the ophthalmology imaging device **20** selected at step S2 to perform an imaging-related operation via the software IF **1013** (step S3, an instruction step).

(103) In the ophthalmology imaging device **20**, the operation control unit **202** executes the imaging-related operation instructed at step S3 (step S4). Thereby, the imaging unit **206** captures the subject's eye image **21**. Then, the operation control unit **202** transmits the subject's eye image **21** to the ophthalmology information processing device **10** (step S5).

(104) The processing in the ophthalmology information processing device **10** is as follows. The

image acquisition unit **1012** acquires the subject's eye image **21** transmitted at step **S5**. More specifically, the image acquisition unit **1012** acquires the eye image **21** from the ophthalmology imaging device **20** via the software IF **1013** (step **S6**, an acquisition step). That is, the image acquisition unit **1012** acquires the subject's eye image **21** that is taken according to the instruction at step **S3**. The image acquisition unit **1012** transmits the subject's eye image **21** acquired at step **S6** to the diagnosis support system **30** (step **S7**).

(105) In the diagnosis support system **30**, the diagnosis support unit **301** receives the subject's eye image **21** acquired at step **S6** and transmitted at step **S7**. The diagnosis support unit **301** receives, as an input, the received subject's eye image **21** and outputs, as an output, the diagnosis support information **31** for the subject's eye image **21** (step **S8**, a diagnosis support step).

(106) In the ophthalmology information processing device **10**, the subject information acquisition unit **104** acquires, from the electronic medical record system **50**, the subject information **53** on the subject whose eye image **21** was taken at step **S6** (step **S9**). Note that step **S9** may be executed before step **S7**.

(107) The association unit **105** associates the diagnosis support information **31** output at step **S8** with the identification information **54** included in the subject information **53** acquired at step **S9**. Thereby, the association unit **105** generates the diagnosis support information **32** (step **S10**). The report generation unit **106** generates the report **41** including the diagnosis support information **32** generated at step **S10**. The report generation unit **106** transmits the generated report **41** to the electronic medical record system **50** (step **S11**).

(108) The electronic medical record system **50** acquires the report **41** transmitted at step **S11** (step **S12**). The electronic medical record system **50** transmits the acquired report **41** to the output device **40** (step **S13**). The output device **40** acquires the report **41** transmitted at step **S13**. The output device **40** outputs the acquired report **41** (step **S14**).

(109) Through the above procedure, the ophthalmology information processing system **1** acquires the subject's eye image **21** taken by the ophthalmology imaging device **20** operating in the remote mode. The ophthalmology information processing system **1** outputs the diagnosis support information **31** regarding the subject's eye image **21** based on the acquired subject's eye image **21**. Further, the ophthalmology information processing system **1** generates the report **41** including the diagnosis support information **32**. The ophthalmology information processing system **1** transmits the generated report **41** to the electronic medical record system **50**. Then, the electronic medical record system **50** transmits the report **41** to the output device **40**. Thereby, the subject or a medical person who is the user of the output device **40** can view the report **41**.

(110) Further, the software IF **1013** is used for communication with the ophthalmology imaging device **20**. Therefore, the ophthalmology information processing system provider can easily design and develop software required for interacting with the ophthalmology imaging device **20** at low cost. In addition, the ophthalmology information processing system provider can design and develop the specifications and software for the ophthalmology imaging device **20** with a high degree of freedom without subject to constraints such as the environment.

(111) Furthermore, the ophthalmology information processing system **1** can generate the diagnosis support information **32** associated with the identification information **54** without depending on the ophthalmology imaging device **20**. Therefore, it is not necessary to register and manage identification information of subjects in the ophthalmology imaging device **20**. Furthermore, the identification information does not need to be linked between the ophthalmology imaging device **20** and the ophthalmology information processing system **1**. For the reasons described above, using the ophthalmology information processing system **1**, it is possible to reduce the complexity involved in registering and managing the identification information of subjects.

(112) FIG. **10** is a sequence diagram showing an example of a flow of processing until the report **41** is obtained based on the subject's eye image **22** taken by the ophthalmology imaging device **20** operating in the local mode. Each process in the ophthalmology information processing system **1** is

executed and controlled by one or more information processing devices. At least one information processing device may be communicably connected to the ophthalmology imaging device **20**.

(113) In the ophthalmology imaging device **20**, the mode switching unit **201** switches the mode to the local mode in response to an input operation received by the input unit **207** (step S21). The operation control unit **202** executes the imaging-related operations according to the input operation from the input unit **207** (step S22). The subject information receiving unit **203** receives input or selection of the subject information **23** including at least the identification information **24** in accordance with the input operation from the input unit **207** (step S23). Step S23 may be executed before step S22. The association unit **205** generates the subject's eye image **21** taken by the imaging-related operation of the by the imaging unit **206** executed at step S22. Furthermore, the association unit **205** generates the subject's eye image **22** based on the subject information **23** received at step S23 (step S24). The operation control unit **202** transmits the subject's eye image **22** generated at step S24 to the ophthalmology information processing device **10** (step S25).

(114) In the ophthalmology information processing device **10**, the image acquisition unit **1012** acquires the subject's eye image **22** transmitted at step S25 via the software IF **1013** (step S26). The image acquisition unit **1012** transmits the subject's eye image **22** acquired at step S26 to the diagnosis support system **30** (step S27).

(115) In the diagnosis support system **30**, the diagnosis support unit **301** receives the subject's eye image **22** transmitted at step S27. The diagnosis support unit **301** inputs the received eye image **22**. Then, the diagnosis support unit **301** outputs the diagnosis support information **33** for the subject's eye image **22** (step S28).

(116) The report generation unit **106** generates the report **41** including the diagnosis support information **33** output at step S28. The report generation unit **106** transmits the generated report **41** to the electronic medical record system **50** (step S29).

(117) The electronic medical record system **50** acquires the report **41** transmitted at step S29 (step S30). The electronic medical record system **50** transmits the acquired report **41** to the output device **40** (step S31). The output device **40** acquires the report **41** transmitted at step S31. Then, the output device **40** outputs the report **41** (step S32).

(118) Through the above procedure, the ophthalmology information processing system **1** acquires the subject's eye image **22** taken by the ophthalmology imaging device **20** operating in the local mode. The ophthalmology information processing system **1** outputs the diagnosis support information **33** regarding the subject's eye image **22** based on the acquired subject's eye image **22**. Further, the ophthalmology information processing system **1** generates the report **41** including the diagnosis support information **33**. The ophthalmology information processing system **1** transmits the generated report **41** to the electronic medical record system **50**. Then, the electronic medical record system **50** transmits the report **41** to the output device **40**. Thereby, the subject or a medical person who is the user of the output device **40** can view the report **41**.

(119) In this way, the ophthalmology information processing system **1** can also generate the diagnosis support information **33** and the report **41** based on the subject's eye image **22** taken by the ophthalmology imaging device **20** operating in the local mode.

(120) The ophthalmology information processing system **1** may be provided by the ophthalmology information processing system provider as a program for realizing each function of the ophthalmology information processing system **1**. That is, the ophthalmology information processing system provider may provide a program for realizing each function of the operation instruction unit **1011** that executes the instruction step, the image acquisition unit **1012** that executes the acquisition step, and the diagnosis support unit **301** that executes the diagnosis support processing step. At the instruction step, instructions are generated in accordance with the specifications of the software IF **1013**. At the instruction step, the instruction is given to the ophthalmology imaging device **20**. The ophthalmology imaging device **20** executes the imaging-related operations based on the instruction. Therefore, each function realized by the program

provided by the ophthalmology information processing system provider is used. As a result, a series of processes from taking the subject's eye image **21** by the ophthalmological photographing device **20** to obtaining the diagnosis support information **31** and **33** based on the subject's eye image **21** can be performed without being constrained. This restriction depends on other specifications of the ophthalmology imaging device **20** provided by the ophthalmology imaging device provider, the environment, and the like. Further, for the ophthalmology imaging device provider, there is no need to customize the ophthalmology imaging device **20** to meet the specifications of the ophthalmology information processing system **1** provided by the ophthalmology information processing system provider. Accordingly, the ophthalmology information processing system provider can efficiently operate the ophthalmology imaging device **20** based on instructions from the ophthalmology information processing system **1**.

(121) In the present disclosure, the term “processor” may refer to a single hardware processor or several hardware processors that are configured to execute computer program code (i.e., one or more instructions of a program) included in a program. In other words, a processor may be one or more programmable hardware devices. For instance, a processor may be a general-purpose or embedded processor and include, but not necessarily limited to, CPU (a Central Processing Circuit), a microprocessor, GPU (a Graphics Processing Unit), and DFP (a Data Flow Processor).

(122) The term “memory” in the present disclosure is a non-transitory, tangible storage medium, and may refer to a single or several hardware memories configured to store computer program code (i.e., one or more instructions of a program) and/or data accessible by a processor. A memory may be implemented using any suitable memory technology, such as static random-access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. Computer program code constituting a program may be stored on the memory and, when executed by a processor, cause the ophthalmology information processing system to perform the above-described various functions.

(123) In the present disclosure, the term “circuit” may refer to a single hardware logical circuit or several hardware logical circuits (in other words, “circuitry”) that are configured to cause the ophthalmology information processing system to perform one or more functions. In other words (and in contrast to the term “processor”), the term “circuit” refers to one or more non-programmable devices. For instance, a circuit may be a customized IC (an Integrated Circuit) that is customized for a particular use and is configured as non-programmable.

(124) In the present disclosure, the phrase “at least one of (i) a circuit and (ii) a processor” should be understood as disjunctive (logical disjunction) where the circuit and the processor can be optional and not be construed to mean “at least one of a circuit and at least one of a processor”. Therefore, in the present disclosure, the phrase “at least one of a circuit and a processor is configured to cause the ophthalmology information processing system to perform functions” should be understood as “only the circuit can cause the ophthalmology information processing system to perform all the functions”. Further, the phrase “at least one of a circuit and a processor is configured to cause the ophthalmology information processing system to perform functions” should be understood as “only the processor can cause the ophthalmology information processing system to perform all the functions”. Moreover, the phrase “at least one of a circuit and a processor is configured to cause the ophthalmology information processing system to perform functions” should be understood as “the circuit can cause the ophthalmology information processing system to perform at least one of the functions and the processor can cause the ophthalmology information processing system to perform the remaining functions”. In the last example, if the ophthalmology information processing system performs functions A to C, for example, the functions A and B among the functions A to C may be implemented via a circuit, while the remaining function C may be implemented via a processor.

(125) The present disclosure is not necessarily limited to the embodiments described above.

Claims

1. A control method executed by an information processing device communicably connected to an ophthalmology imaging device including (i) an imaging unit that is configured to take a subject's eye image and (ii) an operation control unit that is configured to execute an imaging-related operation related to taking of the subject's eye image in accordance with an input operation directly input into the ophthalmology imaging device and is configured to execute the imaging-related operation in accordance with an instruction remotely transmitted, the method comprising: an instruction step of transmitting the instruction to the ophthalmology imaging device as an operation target to execute the imaging-related operation; an acquisition step of acquiring, from the ophthalmology imaging device, the subject's eye image that was taken in accordance with the instruction; and a diagnosis support step of outputting diagnosis support information for the subject's eye image, wherein the imaging-related operation includes at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of the imaging unit, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state, the instruction step further comprises generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface, the ophthalmology imaging device is provided by a first business provider, a program for causing the information processing device to execute the instruction step, the acquisition step, and the diagnosis support step is provided by a second business provider that is different from the first business provider, the instruction step further comprises generating the instruction to meet the specification of the API or the SDK that was provided by the first business provider to the second business provider, and the operation control unit executes the imaging-related operation in accordance with the instruction.
2. The control method according to claim 1, wherein the imaging-related operation further includes transfer of imaging data.
3. A control method executed by an information processing device communicably connected to an ophthalmology imaging device including (i) an imaging unit that is configured to take a subject's eye image and (ii) an operation control unit that is configured to execute an imaging-related operation related to taking of the subject's eye image in accordance with an input operation directly input into the ophthalmology imaging device and is configured to execute the imaging-related operation in accordance with an instruction remotely transmitted, the method comprising: an instruction step of transmitting the instruction to the ophthalmology imaging device as an operation target to execute the imaging-related operation; an acquisition step of acquiring, from the ophthalmology imaging device, the subject's eye image that was taken in accordance with the instruction; and a diagnosis support step of outputting diagnosis support information for the subject's eye image, wherein the imaging-related operation includes at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of the imaging unit, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state, the instruction step further comprises generating the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface, the ophthalmology imaging device includes a mode switching unit that is configured to switch an operation mode of the ophthalmology imaging device between a local mode and a remote mode, the operation control unit executes the imaging-related operation in accordance with the input operation directly input into the ophthalmology imaging device unit during the local mode, and the operation control unit executes the imaging-related operation in accordance with the instruction remotely transmitted from the information processing device during the remote mode the

ophthalmology imaging device comprises: a subject information receiving unit that is configured to receive identification information of a subject during the local mode; and an association unit that is configured to associate the subject's eye image taken by the ophthalmology imaging device with the identification information received by the subject information receiving unit, and the operation control unit is configured to: transmit the subject's eye image being associated with the identification information to the information processing device during the local mode; and transmit the subject's eye image not being associated with the identification information to the information processing device during the remote mode.

4. The control method according to claim 3, wherein when the ophthalmology imaging device is in the remote mode, the operation control unit is configured to execute the imaging-related operation regardless of whether the identification information is received.

5. An ophthalmology information processing system communicably connected to an ophthalmology imaging device including (i) an imaging unit that is configured to take a subject's eye image and (ii) an operation control unit that is configured to execute an imaging-related operation related to taking of the subject's eye image in accordance with an input operation directly input into the ophthalmology imaging device and is configured to execute the imaging-related operation in accordance with an instruction remotely transmitted, the system comprising: at least one of (i) a circuit and (ii) a processor having a memory storing computer program code, wherein the at least one of the circuit and the processor is configured to cause the ophthalmology information processing system to: transmit the instruction to the ophthalmology imaging device as an operation target to execute the imaging-related operation; acquire, from the ophthalmology imaging device, the subject's eye image that was taken in accordance with the instruction; and output diagnosis support information for the subject's eye image, wherein the imaging-related operation includes at least one of starting of an imaging sequence, stopping of the imaging sequence, switching between a left subject's eye and a right subject's eye, optimization of the imaging unit, alignment, capture, an initial operation associated with switching of patients, and transition to an imaging standby state, the at least one of the circuit and the processor is further configured to cause the ophthalmology information processing system to generate the instruction to meet a specification of an API (Application Programming Interface) or an SDK (Software Development Kit) as a software interface, the ophthalmology imaging device is provided by a first business provider, a program including the computer program code for the ophthalmology information processing system is provided by a second business provider that is different from the first business provider, the at least one of the circuit and the processor is further configured to cause the ophthalmology information processing system to generate the instruction to meet the specification of the API or the SDK that was provided by the first business provider to the second business provider, and the operation control unit of the ophthalmology imaging device executes the imaging-related operation in accordance with the instruction.

6. The control method according to claim 3, wherein the imaging-related operation further includes transfer of imaging data.
